DO NOW — RECALL

Input and output devices

Name: _____ Class: _____ Date: _____

1. Why is a GPU suitable for applying the same filter to many independent pixels?

2. What role does a processor's control unit have?

3. What data type should represent the state of a binary switch?

4. ASCII encodes A as decimal 65 and B as 66. A string 'ABBA' uses four 8-bit stored code units in this simplified example. How many bits and bytes, and why is this not proof all Unicode characters fit in one byte?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It can apply similar computations to many data elements in parallel.

2. Model answer

It decodes and coordinates instruction execution.

3. Model answer

Boolean.

4. Model answer

4×8=32 bits=4 bytes. Unicode contains many more characters, and UTF encodings may use more than one byte for a character; 8-bit storage here is an assumption.

Vocabulary — know these terms

sensor — Input transducer measuring a physical quantity.

ADC — Analogue-to-Digital Converter.

actuator — Output component that causes physical action.

latency — Delay between input and system response.

accessibility — Design usable by people with varied needs.

NEW LEARNING

Read, annotate and explain

Select suitable input

A temperature sensor measures a physical quantity; an analogue-to-digital converter is needed if the controller uses digital samples.

Process and control

Software compares the sampled temperature with a target to decide whether to open the vent.

Choose appropriate outputs

An actuator makes the physical change; a display can show readings. Design must address accessibility and suitable latency.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Choose input and output for an automatic door that opens when someone approaches.

Predict first

A proximity or motion sensor detects someone approaching and passes a signal to the controller.

Explain and check

A motor and driver create movement; safety sensing reduces the risk of closing onto a person.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Name a suitable input and output device for a weather station, and explain the role of each.

Question 2

A greenhouse opens a vent when temperature exceeds a limit. Name the sensor, processing decision and actuator, and explain any analogue conversion.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate device choices for a pedestrian crossing serving users with hearing or visual impairment.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A motor that physically opens a door is an input device.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.